## a 6502 finite census: all completed arms

| L/BRK/budget | Periods | Words done | Exact |
|---|---|---|---|
| 16/fait/128 | 1,2,3,4 | 4,311,810,304 | 0 |
| 16/fait/512 | 1,2,3,4 | 4,311,810,304 | 0 |
| 16/halt/128 | 1,2,3,4 | 4,311,810,304 | 0 |
| 16/halt/512 | 1,2,3,4 | 4,311,810,304 | 0 |
| 20/fait/128 | 1,2,3 | 16,843,008 | 0 |
| 20/fait/512 | 1,2,3 | 16,843,008 | 0 |
| 20/halt/128 | 1,2,3 | 16,843,008 | 0 |
| 20/halt/512 | 1,2,3 | 16,843,008 | 0 |
| 32/fait/128 | 1,2,3 | 16,843,008 | 0 |
| 32/fait/512 | 1,2,3 | 16,843,008 | 0 |
| 32/halt/128 | 1,2,3 | 16,843,008 | 0 |
| 32/halt/512 | 1,2,3 | 16,843,008 | 0 |

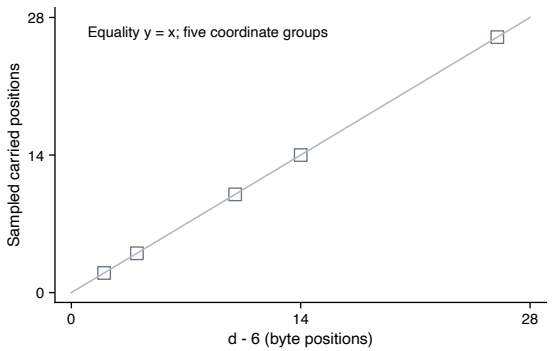
L = tape bytes; budget = CPU instructions; k = tiled-word period
fait = faithful BRK; halt = halt before BRK; Exact = self-writers
Words done sums k; each L16 k4 arm alone exhausts $256^4$ words

## b 8086 finite census: declared reset

| L/budget/k | Words | Zero exact | ≥75% ≥8/16 |
|---|---|---|---|
| 16/128/1 | 256 | 0 | 0 |
| 16/128/2 | 65,536 | 20 | 29 |
| 16/512/1 | 256 | 0 | 0 |
| 16/512/2 | 65,536 | 29 | 29 |
| 32/128/1 | 256 | 0 | 0 |
| 32/128/2 | 65,536 | 26 | 36 |
| 32/512/1 | 256 | 0 | 0 |
| 32/512/2 | 65,536 | 26 | 25 |

L bytes; budget instructions; k = word period; SP at ring end
Counts of nonzero words: exact output into zero partner; or
≥75% byte match in ≥8 of 16 random-partner encounters

## c Constructed 6502: sampled carriage




296 constructed copier/contract configurations; points coincide
Coordinate counts 2:96, 4:40, 10:96, 14:32, 26:32; d offset bytes
Site: ≥half tested alternatives heritable and allele retained ≥50%

## d S1: complete four-arm route ledger

| Arm L/budget | n | Conf. | Loop | Open | Undet. |
|---|---|---|---|---|---|
| A 16/128 | 20 | 20 | 14 | 0 | 0 |
| B 32/128 | 20 | 4 | 4 | 0 | 16 |
| C 16/512 | 20 | 19 | 15 | 1 | 0 |
| D 16/128 | 20 | 18 | 17 | 0 | 2 |

S1 = 6502 worlds; Conf. = zero partner fetch; Open = ≥8/16 traces
Undet. = no heritable copier detected; Loop overlaps Conf. (PC return)
20 worlds/arm; L bytes / CPU budget; A–C faithful BRK, D pre-halt

## e Registered C2: complete sampling roles

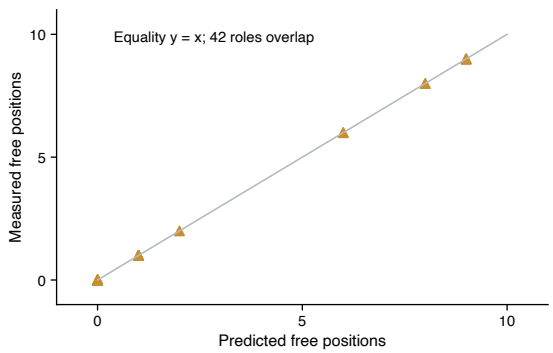
| Sampling role | Sampled | Eligible | Elig. worlds |
|---|---|---|---|
| Final random cell | 640 | 23 | 4 |
| Final top-3 by abundance | 240 | 10 | 4 |
| First detected tape | 62 | 9 | 9 |

**Eligible: 42 roles; 41 unique tapes; 11 worlds**

Pointer randomization: unmeasured (not an S2 arm)

Eligible = certified fixed-core, pure forward-copy family
942 roles; world counts overlap across roles (do not sum)

## f C2 exact capacities: all eligible roles




Free byte positions: arbitrary alternatives retained with causal roots
Exact-copy/core/ancestry mismatches: 0/91,428 direct cases
and 0/731,424 serial observations; 42 roles /41 tapes /11 worlds